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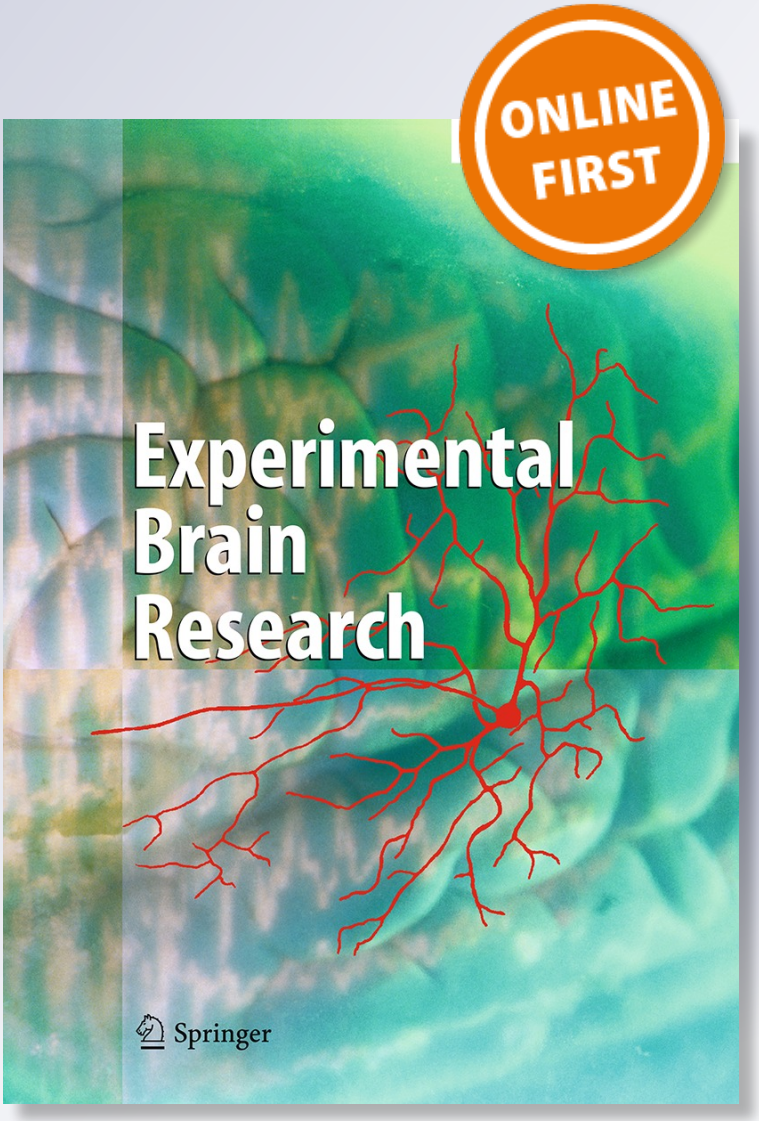


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RESEARCH ARTICLE

# Structural differences in basal ganglia of elite running versus martial arts athletes: a diffusion tensor imaging study

**Yu-Kai Chang[1] · Jack Han-Chao Tsai[2] · Chun-Chih Wang[1] · Erik Chihhung Chang[2]**

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**Abstract** The aim of this study was to use diffusion tensor imaging (DTI) to characterize and compare microscopic differences in white matter integrity in the basal ganglia between elite professional athletes specializing in running and martial arts. Thirty-three young adults with sport-related skills as elite professional runners ($n = 11$) or elite professional martial artists ($n = 11$) were recruited and compared with non-athletic and healthy controls ($n = 11$). All participants underwent health- and skill-related physical fitness assessments. Fractional anisotropy (FA) and mean diffusivity (MD), the primary indices derived from DTI, were computed for five regions of interest in the bilateral basal ganglia, including the caudate nucleus, putamen, globus pallidus internal segment (GPi), globus pallidus external segment (GPe), and subthalamic nucleus. Results revealed that both athletic groups demonstrated better physical fitness indices compared with their control counterparts, with the running group exhibiting the highest cardiovascular fitness and the martial arts group exhibiting the highest muscular endurance and flexibility. With respect to the basal ganglia, both athletic groups showed significantly lower FA and marginally higher MD values in the GPi compared with the healthy control group. These findings suggest that professional sport or motor skill training is associated with changes in white matter integrity in specific regions of the basal ganglia, although these positive changes did not appear to depend on the type of sport-related motor skill being practiced.

**Keywords** DTI · Fitness · Globus pallidus · Putamen · Sport mode

## Introduction

Having had years of intensive training, elite athletes exhibit extraordinary motor skill performance. Relative to novices, trained athletes often display a range of superior movement characteristics, including increased accuracy and automaticity of action (Milton et al. 2004) as well as reduced movement variability (Davids et al. 2006). In addition to these behavioral changes, which have been examined in detail over the last two decades (Ericsson 1996), recent studies exploiting emerging neuroimaging techniques have begun to investigate the effects of long-term training on brain plasticity in athletes (Milton et al. 2007; Yarrow et al. 2009).

Using electroencephalograms (EEGs), Hatfield and colleagues observed that, compared with inexperienced controls, athletes show less cortical activation throughout the entire brain as well as less cortico-cortical EEG coherence between the left temporal and midline frontal areas during motor execution—suggesting that cortical processes involved in skill-related tasks require much less effort and are conducted with greater efficiency in athletes (Haufler et al. 2000; Deeny et al. 2003). In addition to changes in brain function, studies using magnetic resonance imaging

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(MRI) have also revealed structural differences in the brain, including increased gray matter in the mid-temporal area, the left posterior intraparietal sulcus, the bilateral frontal, temporal, and cingulate areas in athletes (Wei et al. 2009; Di et al. 2012) or in normal individuals following longitudinal skill training (Driemeyer et al. 2008), providing further evidence for a correlation between motor skill training and structural cortical plasticity.

With respect to motor skill learning and control in normal and sport settings, changes in subcortical structures such as the basal ganglia have garnered particular attention. The basal ganglia are a collection of subcortical nuclei that consist primarily of the caudate nucleus and putamen, together known as the striatum, as well as the globus pallidus, subthalamic nuclei, and substantia nigra (Gerfen and Bolam 2010). The basal ganglia play an essential role in regulating motor behavior and sport skill training, including action selection, motor preparation, and motor skill acquisition (Chakravarthy et al. 2010; Stocco et al. 2010), and they are also involved in habit learning and automaticity (Ashby et al. 2010), learning and memory (Packard and Knowlton 2002), and sequence and category learning (Seger 2006). To examine whether long-term exercise influences the basal ganglia, Chaddock et al. (2010) compared preadolescent children with high and low fitness status and found that children with high fitness also have a larger left caudate nucleus, bilateral putamen, and globus pallidus. A later study that specifically focused on athletes reported that elite basketball players at the university level had larger absolute striatum and corrected-relative striatum volumes compared with healthy control individuals (Park et al. 2011). Considering these results, as well as the link between basal ganglia and motor function, these findings suggest that structural plasticity in the basal ganglia is positively associated with regular exercise and training in sport-related motor skills.

Notably, the types of motor skills acquired by athletes are highly dependent on the specific sports or activities being performed. According to Schmidt and Wrisberg (2008), sport motor skills can be categorized into open skills (e.g., basketball) or closed skills (e.g., jogging, swimming, gymnastics, or routine-based martial arts) based upon environmental predictability. Furthermore, closed skills can be further classified into continuous skills (e.g., jogging), serial skills (e.g., routine-based martial arts), and discrete skills (e.g., shooting) based on task organization. Such categories reveal the diversity of physical actions, movement paradigms, and cognitive demands required by different types of sport-related motor skills, which might lead to different effects on cortical plasticity. By comparing treadmill running to wheel running, Lin et al. (2011) reported that rats in both motor skill groups showed improved performance on tasks that require both the hippocampus and amygdala (e.g., contextual conditioning); however, only rats in the treadmill running group showed improved performance on a task requiring only amygdalar functions (i.e., cued conditioning), consistent with the idea that changes in neuroplasticity are dependent upon the type of trained motor skill. An association between specialized skills and specific brain structures has also been suggested by a number of human studies (May 2011), including a study showing a positive relationship between cerebellar plasticity and skill level in basketball players (Park et al. 2009), as well as a study showing significant differences with respect to motor cortex, corpus callosum, and auditory cortex size between individuals with 15 months of music training and controls (Hyde et al. 2009). However, it remains unclear whether different types of sport-related motor skill training can induce distinct structural differences in the brain.

Along with MRI (Park et al. 2011), EEG (Lalo et al. 2008) and functional MRI (Uchiyama et al. 2012), the use of diffusion tensor imaging (DTI) to examine anatomical and functional changes in the basal ganglia has generated a great deal of attention. DTI is a sophisticated imaging approach to investigate tissue microstructure and integrity by detecting and quantifying the three-dimensional spatial distribution of water diffusion (Basser 1995; Basser and Pierpaoli 1996). In the brain, the motion of water is impeded by barriers such as white matter tracts or parallel myelinated axons, leading to unequal diffusion directionality, also known as anisotropy (Chenevert et al. 1990). Fractional anisotropy (FA) and mean diffusivity (MD) are the predominant indices of neurofibril integration derived from DTI. FA values range from 0 (isotropic diffusion) to 1 (anisotropic diffusion), and it is believed that this metric reflects the strength of axial relative to radial diffusivity. Although it shares the same value range as FA, MD (also known as the apparent diffusion coefficient) is thought to represent the magnitude of diffusion regardless of directionality in a given voxel (Basser and Pierpaoli 1996). In other words, higher MD values indicate a lack of barriers to water diffusion (Le Bihan et al. 2001). Previous DTI studies have employed FA and MD as measures to examine basal ganglia in the context of aging (Wang et al. 2010), Parkinson's disease (Gattellaro et al. 2009), and attention deficit hyperactivity disorder (Silk et al. 2009). However, to the best of our knowledge, researchers have yet to utilize DTI to examine changes in the white matter integrity of basal ganglia due to different sport-related motor skills.

The purpose of the current study was to use DTI to identify differences in microstructural integrity within the basal ganglia between professional athletes involved in running or routine-based martial arts compared with healthy controls. We adopted a region of interest (ROI) approach to

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analyze FA and MD values in subcomponents of the basal ganglia, including the caudate nucleus, putamen, globus pallidus, and subthalamic nucleus. Between-group differences in FA and MD within these ROIs were analyzed. We hypothesized that FA and MD values in both groups of professional athletes would be different than in the control group, and furthermore, we proposed that these metrics would be significantly different between the two athletic groups. FA and MD reflect the directionality of water diffusion—which is determined by the size, shape, and composition of any physical obstructions, as well as the spacing between these obstructions (Beaulieu 2002)—and considering that is unknown how athletic training may affect such variables, we did not have specific hypotheses regarding the directionality of the between-group differences.

## Materials and methods

### Participants

Thirty-three healthy adults ranging in age from 22 to 26 years were recruited and assigned to one of three gender-balanced groups (running, routine-based martial arts, control) based upon self-reported sport history. Participants in the running and martial arts groups were currently active athletes enrolled in either a professional running group (1500 m and marathon) or a professional martial arts group (Wushu, a routine- and performance-based Chinese martial art) from a sport-related university in Taoyuan, Taiwan; participants in the control group were recruited from a non-sport-related university. Participants in both sport groups were characterized as athletes currently receiving an average of 4.2 h of training daily and having engaged in an average of 2.7 h of vigorous training per day for at least 2.8 years. In total, 45 and 9 % of the athletes from the running and martial arts groups, respectively, had received international titles, and all had won at least one national achievement. The control individuals were untrained and had occasionally taken part in sports over the previous 3 years. All participants reported being free of neurological and psychiatric disorders and were right-hand dominant with normal or corrected-to-normal vision. Participants were required to provide written declaration of consent prior to participation. The experimental protocol was in compliance with the Declaration of Helsinki and was approved by the Institutional Review Board. Table 1 shows the details of the participants' sport-related attributes.

### Intelligence measurement

Four subtests from the Wechsler Adult Intelligence Scale-Third Edition (WAIS-III) (Wechsler and Corporation 1997) were employed to evaluate multiple aspects of intelligence in the participants, including the verbal comprehension index (VCI), working memory index (WMI), and perceptual organization index (POI). VCI was assessed using similarity tests, WMI was assessed using digit span and letter-number sequencing, and POI was assessed using picture completion tests.

### Health- and skill-related physical fitness assessments

Health-related physical fitness components, including cardiovascular fitness, muscular strength, muscular endurance, flexibility and body composition, as well as skill-related physical fitness, including agility and power, were examined.

Cardiovascular fitness, a primary component of health-related fitness, has been positively associated with brain morphology and function. Cardiovascular fitness, represented by $VO_{2\ peak}$, was measured with a maximal graded exercise test (GXT) using the Bruce Treadmill Protocol on a motorized treadmill (h/p/cosmos airwalk, Germany) (American College of Sports Medicine 2013).

Muscular strength was determined using averaged handgrip dynamometer values for each hand, and muscular endurance was measured using two of three approaches: 60-s push-ups (men) or 60-s bent knee push-ups (women) along with 30- and 60-s abdominal sit-ups. Flexibility was assessed using the sit-and-reach test to determine lower back and hamstring flexibility. Finally, body composition was calculated using a bioimpedance spectroscopy instrument (InBody 3.0 DS12B887, Dallas) to determine body-mass fat percentage.

With respect to the skill-related physical fitness components, agility and power were evaluated using the *T* test and vertical jump test, respectively (Lustig et al. 2009). The *T* test required running around four cones arranged in a *T* shape as quickly as possible, and the vertical jump test required jumping within a circular area as high as possible.

### Experimental procedure

Participants individually visited two laboratories at different universities on separate days within a 1-week interval. On day one, at the National Taiwan Sport University, participants completed evaluations for demographics, medical and neurological history, and sport history questionnaires to identify each participant's inclusion criteria. Eligible participants were then assessed for intelligence using the four subtests of the WAIS-III, and for health- and skill-related physical fitness using the GXT, handgrip, 60-s push-up, 30- and 60-s abdominal sit-up, sit-and-reach, bioimpedance spectroscopy, *T* test, and vertical jump tests. These testing

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**Table 1** Description of participant demographic and physical data (mean/SD)

| Variable | Group | | | Total ($n = 33$) |
|---|---|---|---|---|
| | Running ($n = 11$) | Martial Art ($n = 11$) | Control ($n = 11$) | |
| Female/male | 2/10 | 3/12 | 5/8 | 10/33 |
| Age (year) | 23.27/1.56 | 23.55/1.21 | 24.00/1.48 | 23.61/1.42 |
| Education (year) | 4.00/0.00 | 3.91/0.30 | 4.00/0.00 | 3.97/0.17 |
| Height (cm) | 171.36/7.22 | 167.73/8.15 | 170.27/8.96 | 169.79/8.03 |
| Weight (kg) | 59.46/6.1 | 66.95/13.41 | 65.27/13.07 | 63.89/11.48 |
| Father's education | 2.33/0.87 | 2.82/0.87 | 3.09/1.05 | 2.77/0.99 |
| Mother's education | 2.67/0.50 | 2.55/0.82 | 3.10/1.37 | 2.77/0.98 |
| Family socioeconomic status | 2.82/0.75 | 3.12/0.98 | 2.73/1.10 | 2.09/1.53 |
| WAIS-III | | | | |
| Similarities (z) | 9.82/2.40 | 1.91/2.47 | 11.18/2.27 | 10.97/2.47 |
| Digit forward span test | 13.18/2.79 | 13.27/2.28 | 13.55/2.11 | 1333/2.34 |
| Digit backward span test | 9.82/2.60 | 10.64/1.96 | 11.64/413 | 10.70/3.04 |
| Digit span test (z) | 9.18/3.02 | 9.91/2.66 | 11.00/3.35 | 10.03/3.03 |
| Letter-number seq. (z) | 8.40/2.84 | 9.64/1.86 | 10.27/2.28 | 9.47/2.40 |
| Picture completion test (z) | 10.50/2.55 | 11.09/4.48 | 12.46/1.44 | 1.36/3.12 |
| Sport characteristics | | | | |
| Training years | 2.82/0.75 | 3.18/0.98 | 0.27/0.65 | 2.09/1.53 |
| Daily training hours | 3.50/0.89 | 4.77/1.01 | 0.14/0.45 | 2.80/2.14 |
| Daily hours of vigorous training | 2.55/0.82 | 2.73/0.79 | 0.10/0.30 | 1.79/1.39 |
| Achievement | | | | |
| International (%) | 45 % | 9 % | – | – |
| National (%) | 100 % | 100 % | – | – |
| Fitness data | | | | |
| $VO_{2\ peak}$ (mL/kg.min) | 70.40/6.96[a] | 53.15/7.42[b] | 43.05/916 | 55.07/13.68 |
| Muscular strength | 72.92/13.82 | 82.29/24.37 | 74.55/23.70 | 76.59/20.93 |
| Muscular endur./press up | 15.82/7.60[a] | 22.82/10.88[a] | 8.12/7.74 | 15.61/10.52 |
| Muscular endur./CCU 30 | 23.73/2.24[a] | 27.55/3.39[b] | 19.36/2.66 | 23.55/4.35 |
| Muscular endur./CCU 60 | 46.64/5.16[a] | 51.82/5.30[b] | 35.36/5.39 | 44.61/8.65 |
| Flexibility (cm) | 37.82/8.34 | 50.86/6.00[a] | 31.50/9.49 | 40.06/11.32 |
| % Body fat mass | 13.93/2.56[a] | 19.49/4.14[a] | 21.75/6.79 | 18.68/5.77 |
| Agility (ms) | 11.87/0.93[a] | 11.20/1.27[a] | 13.33/1.12 | 12.14/1.40 |
| Power (cm) | 54.09/6.88[a] | 60.82/8.69[a] | 48.82/8.99 | 53.58/10.12 |

Letter-number Seq. = letter-number sequencing (z); training years, number of years engaged in sport training; WAIS-III: Wechsler Adult Intelligence Scale, 3rd ed.; and muscular endur./press ups/CCU 30/CCU 60, muscular endurance/press ups, crunch curl-ups for 30 and 60 s, respectively

[a] Represents a significant difference relative to the control group

[b] Represents an additional significant difference between the running and control groups

procedures were administered by investigators with trained psychometrical and fitness testing backgrounds.

On day two, at the National Yang-Ming University, participants underwent standard MRI brain scanning in the MRI examination room. Participants were compensated approximately $20 and $17 US dollars for days one and two, respectively. They were then briefly informed as to the purpose and expectations of the current research at the end of the MRI scanning procedure.

## MRI acquisition and image processing

We performed diffusion tensor imaging (DTI) using a Siemens 3T MAGNETOM TIM Trio scanner (Erlangen, Germany). T1-weighted structural images were acquired using the MPRAGE sequence (TR: 2530 ms; TE: 3.03 ms; flip angle = 7; matrix size: 224 × 256; field of view: 224 × 256 mm; in-plane resolution: 1 × 1 mm; slice thickness: 1.0 mm; 192 slices).

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The DTI scans were acquired using a single-shot echo-planar imaging sequence (TR: 11,000 ms; TE: 104 ms; flip angle = 90°; matrix size: 128 × 128; field of view: 230.4 × 230.4 mm; voxel size: 2 × 2 × 2 $mm^3$; 70 axial oblique slices with 2-mm gaps; 30 non-collinear directions with *b*-values of 1000 s/$mm^2$, and a non-diffusion-weighted image volume with *b*-value 0 s/$mm^2$). Scanning in each direction was repeated three times to increase the signal-to-noise ratio. Due to technical reasons, DTI scans were not performed on one participant in each group, leaving 10 participants in the control, running, and martial arts groups.

The Brain Voyager QX software package (version 2.8.2; Brain Innovation, Maastricht, The Netherlands) and the Neuroelf toolbox (http://neuroelf.net/) were used to preprocess the raw diffusion-weighted images and to compute the FA and MD images using the following steps for each individual participant. First, non-diffusion-weighted ($b = 0$) images were coregistered to their *T*1-weighted images based on least squares as the cost function, followed by visual inspection to ensure the quality of alignment. The same parameters of coregistration were applied to the diffusion-weighted images.[1] Second, the *T*1-weighted images were normalized into Talairach space using a nine-parameter rigid-body affine transformation. The inverse matrix of the transformation was then applied to a standardized template of the basal ganglia to align the regions of interest from the Talairach space to each individual participant's native space. The basal ganglia mask encompassed the caudate nucleus (CAU), putamen (PUT), globus pallidus internal segment (GPi), globus pallidus external segment (GPe), and subthalamic nucleus (STN). The mask was based on a template for the basal ganglia (BGHAT, downloadable from http://lrnlab.org/; Prodoehl et al. 2008) that was manually drawn on the Talairach atlas, and it was highly consistent with the individually drawn ROIs. Homologous basal ganglia subregions in both the left and right hemispheres were selected and spatially transformed into each individual participant's native space, resulting in five pairs of basal ganglia ROIs for each participant. To avoid the partial volume problem caused by averaging with surrounding tissues, the centroids of the ROIs in each hemisphere were computed, and new spherical ROIs (2 mm radius) were created surrounding the centroids of the template ROIs. Visual inspection of these new spherical ROIs ensured that they were all located within the designated basal ganglia regions. Third, estimation of diffusion tensors was performed using the 3D diffusion-weighted volumes. Individual participant's FA and MD values at each voxel were then computed for each pair of template and spherical ROIs and then subjected to group analysis.

[1] Because the latest version of Brain Voyager QX (v2.8.2) does not yet provide tools for Eddy current compensation and motion correction, the analysis reported here did not include these steps.

### Statistical analysis

A cross-sectional design was employed in the present study. One-way between-subjects analysis of variance (ANOVA) or independent Kruskal–Wallis tests were conducted to analyze differences in the demographic and physical characteristics between the three groups (control, running, and martial arts) where appropriate (i.e., continuous and categorical variables). Tukey's HSD post hoc test was used when significant differences between groups were detected.

Two-way group × hemisphere (left/right) mixed-design ANOVAs were applied to the averaged FA and MD values within each template or spherical ROI for the three groups of participants. To achieve a balance between precision in locating ROIs and representativeness of the data, only ROIs that were significant for both template and spherical cases were considered valid and reported. Post hoc comparisons were conducted using Tukey's HSD based on the results from the less significant ROI in the between-subjects ANOVA.

## Results

### Participant demographic data

Table 1 shows the details of the participants' demographic and physical fitness data. There were no significant differences in age, education, height, or weight between the three groups [$F(2, 31) = 0.39$–$1.52$, $p = 0.28$–$0.68$], nor were there significant differences in gender, parental education, or family socioeconomic status between the three groups (Kruskal–Wallis test, $p = 0.21$–$0.74$).

Similarly, there were no significant differences between the three groups on the similarity, digit forward span, digit backward span, letter-number sequencing, or picture completion tests of the WAIS-III [$F(2, 31) = 0.38$–$1.71$, $p = 0.19$–$0.94$]. In other words, we found no significant differences in the demographics or intelligence of the participants across the three groups.

### Health- and skill-related physical fitness data

We observed significant differences in $VO_{2\ peak}$, push-ups, sit-ups for 30 or 60 s, flexibility, percent body fat mass, agility, or power [$F(2, 31) = 6.41$–$31.61$, $p$ values $=0.001$–$0.005$]. No differences were observed for handgrip strength.

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**Table 2** Average FA values in each of the five template and spherical basal ganglia ROIs across the three groups

| Region | Running | Martial art | Control |
|---|---|---|---|
| *Caudate* | | | |
| Template | 0.3340(0.0376) | 0.3256(0.0565) | 0.3671(0.0573) |
| Spherical | 0.2133(0.0330) | 0.2113(0.0469) | 0.2696(0.1046) |
| *Putamen* | | | |
| Template | 0.3182(0.0172) | 0.3266(0.0242) | 0.3400(0.0263) |
| Spherical | 0.1434(0.0247) | 0.1573(0.0641) | 0.1672(0.0403) |
| *GPi* | | | |
| Template | 0.3673(0.0302) | 0.3635(0.0317) | 0.4082(0.0537) |
| Spherical | 0.2962(0.0304) | 0.2710(0.0446) | 0.3463 (0.0470) |
| *GPe* | | | |
| Template | 0.2367(0.0231) | 0.2399(0.0245) | 0.2857(0.0449) |
| Spherical | 0.2037(0.0162) | 0.1981(0.0107) | 0.2169(0.0162) |
| *STN* | | | |
| Template | 0.4775(0.0300) | 0.4893(0.0615) | 0.5133(0.0726) |
| Spherical | 0.4883(0.0425) | 0.5163(0.0777) | 0.5194(0.0852) |

Values in parentheses indicate the standard deviation

*GPi* globus pallidus internal segment, *GPe* globus pallidus external segment, *STN* subthalamic nucleus

The follow-up post hoc comparison revealed that both sport groups had significantly better fitness with respect to push-ups, percentage body fat mass, agility, and power compared with the control group (*p*s value <0.05). The running group showed the highest $VO_{2\ peak}$, followed by the martial arts group (*p* value <0.01), and then the control group (*p* value <0.001). Additionally, the martial arts group showed the highest fitness, including sit-ups for 30 and 60 s, followed by the running group (*p*s value <0.03), and then the control group (*p* value = 0.01). The martial arts group also performed best on the flexibility test (*p* value <0.002).

These data indicate that individuals in the sport groups generally had better fitness than those in the control group; the running group showed the highest $VO_{2\ peak}$, and the martial arts group had the best sit-ups for 30 and 60 s and flexibility scores (Table 1).

### DTI measures

Table 2 shows the average FA values of the five pairs of template and spherical ROIs for each group. Among all ROIs, only averaged FA values in the GPi [$F_{spherical}(2, 27) = 6.063$, $p = 0.007$; $F_{template}(2, 27) = 3.854$, $p = 0.03$] showed significant differences between the groups. Post hoc comparisons showed that the martial arts group ($p_{template} = 0.047$; $p_{spherical} = 0.005$) and the running group ($p_{template} = 0.076$; $p_{spherical} = 0.077$) had significantly or marginally lower FA values, respectively, than the control group. The difference between the running and martial arts groups was not significant (Fig. 1). The hemisphere effects were significant for both types of ROIs only in GPe [$F_{spherical}(1, 27) = 9.674$, $p = 0.004$; $F_{template}(1, 27) = 5.105$, $p = 0.032$) and STN ($F_{spherical}(1, 27) = 14.723$, $p < 0.001$; $F_{template}(2, 27) = 22.978$, $p < 0.001$]. The average FA was higher in the left (0.23) than in the right GPe (0.18), and higher in the right (0.54) than in the left STN (0.48). Because no interaction in any ROI approached significance (see Supplementary Table 1) and lateralization is not of theoretical interest in the current study, the effect of hemisphere would not be further addressed.

The average MD values of the five ROIs for each group are shown in Table 3. Average MD values were only significantly different between the groups for GPi [$F_{spherical}(2, 27) = 3.71$, $p = 0.04$; $F_{template}(2,27) = 6.348$, $p = 0.006$]. Post hoc comparisons revealed marginally higher MD values in the running and martial arts groups compared with the control group ($p = 0.06$ and 0.07, respectively, Fig. 1). The effects of hemisphere were significant for both types of ROIs only in caudate [$F_{spherical}(1, 27) = 10.525$, $p = 0.003$; $F_{template}(1, 27) = 7.525$, $p = 0.01$) and GPi ($F_{spherical}(1, 27) = 10.886$, $p = 0.003$; $F_{template}(1, 27) = 12.151$, $p = 0.005$]. The average MD was higher in the left (0.31) than in the right (0.29) caudate, and higher in the right GPi (0.23) than in the left STN (0.22). Likewise to FA, no interaction reached significance for both types of ROIs (see Supplementary Table 2).

## Discussion

The current study builds upon previous reports that indicated a positive relationship between athletic activity and brain plasticity by showing differences in white matter integrity in the basal ganglia between professional athletes involved in intensive running and martial arts training and normal control individuals. Participants from each group were recruited based on previous sport experience and achievement, and no significant differences were found between the groups in terms of demographic characteristics such as age, education, or intelligence. Additionally, although athletes in both sport groups had better physical fitness indices in general, athletes in the running group exhibited the best cardiovascular fitness, and athletes in the martial arts group generally showed better values for all other physical fitness indices (e.g., muscular endurance and flexibility). These findings suggest that different sport groups have specialized motor skills and that our group assignments were appropriate.

Regarding DTI measures, we found significant differences between the athlete groups and their control counterparts, but not between the running and martial arts groups,

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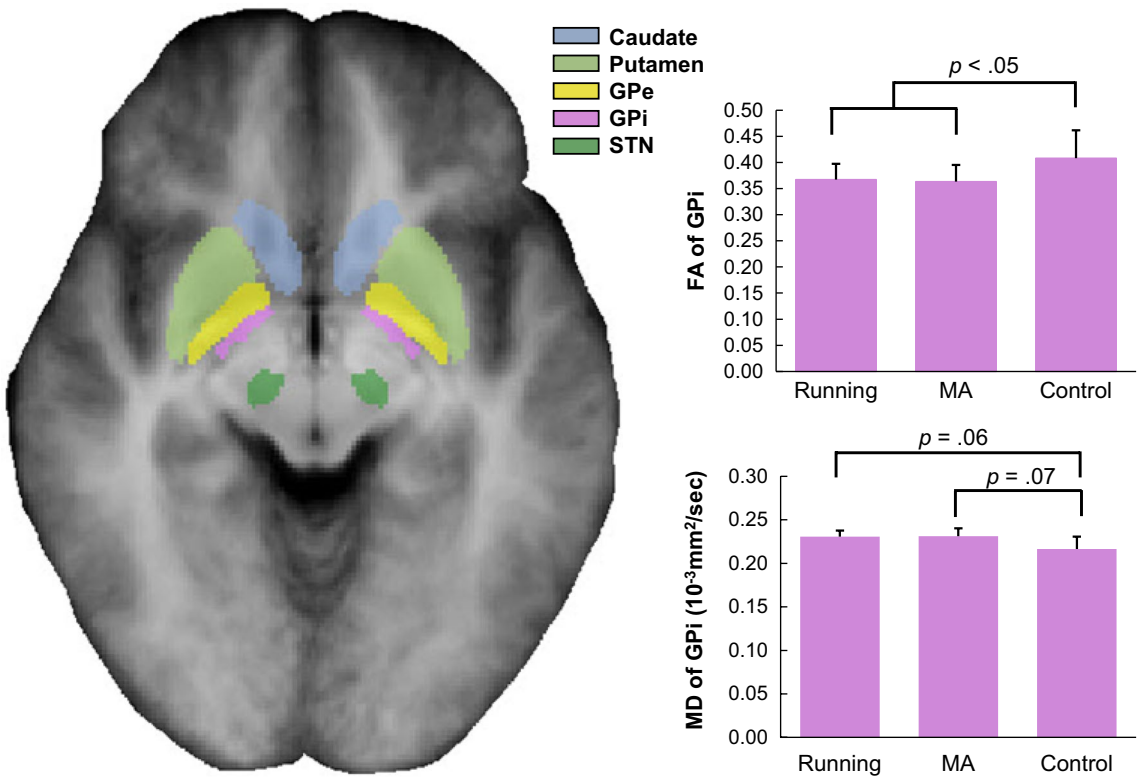


**Fig. 1** Basal ganglia ROIs and significant FA/MD measures. *Bar graphs* were plotted for the ROI with statistical differences between three groups in FA or MD values from both spherical and template methods (only template values were shown here). *Error bars* represent the standard deviation. *MA* martial art group, *GPi* globus pallidus internal segment, *GPe* globus pallidus external segment, *STN* subthalamic nucleus

**Table 3** Average mean diffusivity (MD; $10^{-3}$mm$^2$/s) values in each of the five template and spherical basal ganglia ROIs across the three groups

| Region | Running | Martial art | Control |
|---|---|---|---|
| *Caudate* | | | |
| Template | 0.2935(0.0268) | 0.3019(0.0434) | 0.3108(0.0482) |
| Spherical | 0.2746(0.0885) | 0.2577(0.0629) | 0.3072(0.0956) |
| *Putamen* | | | |
| Template | 0.2287(0.0040) | 0.2270(0.0053) | 0.2241(0.0064) |
| Spherical | 0.2108(0.0079) | 0.2140(0.0063) | 0.2039(0.0079) |
| *GPi* | | | |
| Template | 0.2306(0.0070) | 0.2312(0.0091) | 0.2164(0.0142) |
| Spherical | 0.2273(0.0101) | 0.2285(0.0105) | 0.2095(0.0241) |
| *GPe* | | | |
| Template | 0.2207(0.0038) | 0.2196(0.0052) | 0.2146(0.0097) |
| Spherical | 0.2134(0.0162) | 0.2131(0.0107) | 0.2043(0.0162) |
| *STN* | | | |
| Template | 0.2205(0.0054) | 0.2195(0.0064) | 0.2196(0.0109) |
| Spherical | 0.2188(0.0060) | 0.2187(0.0069) | 0.2198(0.0168) |

Values in parentheses indicate the standard deviation

*GPi* globus pallidus internal segment, *GPe* globus pallidus external segment, *STN* subthalamic nucleus

suggesting that athletic experience in general, but not specific sports, influence the basal ganglia. Specifically, both groups of professional athletes demonstrated lower FA values in the bilateral GPi compared with the controls. By contrast with the results for FA, both athlete groups showed marginally higher MD values than the controls for bilateral GPi, though the effect was not significant. Structurally, the globus pallidus sends tonic inhibitory projections to the ipsilateral motor thalamus via its internal segment (the GPi), as well as to the subthalamic nucleus, ventral anterior (VA), and ventrolateral (VL) nuclei via its external segment (the GPe). These projections primarily consist of GABAergic neurons (Gerfen and Bolam 2010). The GPi and GPe belong to the direct and indirect pathways, respectively, whose relative activity levels determine the tonic inhibitory pressure on the VA and VL. Excitatory projections from the VA and VL to the premotor cortex are involved in planning and initiating movements. Although the interplay between all basal ganglia subcomponents determines the level of excitation in the premotor cortex, our findings indicate that chronic and high-intensity training in skillful movements, regardless of the sport or motor skills involved, can affect the output regions of the basal ganglia.

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Our findings of lower FA and marginally higher MD values in professional athletes compared with control individuals are partially consistent with previous studies examining the relationship between behavioral expertise and FA/MD values (Schmithorst and Wilke 2002; Bengtsson et al. 2005; Imfeld et al. 2009; Hänggi et al. 2010). Hänggi et al. (2010) showed that, relative to non-dancers, professional ballet dancers had reduced gray matter volume in the putamen as well as decreased white matter volume in the corticospinal tract, bilateral internal capsules, corpus callosum, and left anterior cingulum. Similarly, a positive correlation has been found between childhood/adolescent piano practicing time and FA values in a variety of related brain regions (Bengtsson et al. 2005; cf. Han et al. 2009), and there are reports of lower FA and higher MD values in musicians compared with controls (also see Schmithorst and Wilke 2002; Imfeld et al. 2009). These findings suggest that expertise in a specialized motor skill may be associated with changes in microstructural plasticity in the basal ganglia.

Compared with neuroimaging methods that analyze macroscopic volume information (e.g., voxel-based morphometry), DTI measurements can reveal microscopic structural properties of the brain (Hasan et al. 2009). FA is a measurement of diffusional anisotropy and therefore reflects microstructural "directionality," whereas MD is sensitive to changes in diffusion magnitude and therefore reflects microstructural "viscosity." Furthermore, MD measurements can detect non-specific changes in water diffusion restriction (Gong et al. 2014). In the current literature, between-group differences in diffusivity measures in the deep gray matter are studied more frequently in individual patients than in subgroups of the healthy population. It has been reported that the globus pallidus in elderly individuals showed increased MD and decreased FA values (Bhagat and Beaulieu 2004; Pfefferbaum et al. 2010; Gong et al. 2014). Patients with multiple sclerosis showed different diffusivity properties compared with normal controls in the caudate (Ciccarelli et al. 2001; Hasan et al. 2009; Hannoun et al. 2012), putamen (Ciccarelli et al. 2001), or thalamus (Ciccarelli et al. 2001; Hasan et al. 2011; Hannoun et al. 2012). Finally, it has also been observed that patients with Huntington's disease (Douaud et al. 2005) or spina bifida (Hasan et al. 2008) show elevated diffusion anisotropy. Although many such observations have been made, interpretations of DTI measures in gray matter regions such as the basal ganglia or other deep brain nuclei are not as straightforward as for white matter structures.

A multitude of factors likely interact to alter diffusivity in deep gray matter nuclei, including cell type (Percheron et al. 1984), iron accumulation (Rulseh et al. 2013), and axonal properties. It should be noted, however, that the following hypotheses are speculative. Our results indicated significantly reduced FA and marginally elevated MD values in the GPi of the athlete groups. Specific aspects of running and martial arts may influence the output segments of the basal ganglia and globus pallidus, stimulating microstructural changes by expanding axonal diameter and enhancing radial diffusivity, which could in turn lead to reduced FA and increased MD values (Beaulieu 2002). The globus pallidus consists of large disk-shaped neurons that are perpendicular to the myelinated axons (Percheron et al. 1984), and it is also one of the structures in the brain with the highest iron levels (Haacke et al. 2005). Iron atoms are stored by the protein ferritin, which has a large spherical protein coat. Myelinated axons, large cells, and ferritin protein can together form complex microstructures that modulate the free diffusion of water molecules. Long-term and intense training could result in iron deficiency in elite athletes (Wishnitzer et al. 1983; Kaiser et al. 1989; Chatard et al. 1999), and therefore, lower ferritin levels in the brain may reduce obstructions to water diffusivity, leading to lower FA and higher MD values. This possibility warrants future investigation to systematically compare DTI measures in athletes with and without iron deficiency.

Although our findings extend the previous studies by demonstrating the modulatory effects of behavioral expertise on macro- and microneurofibril structures, some caveats are worth considering. Due to the limitations of cross-sectional experiments, our study did not address changes relative to the baseline levels of FA/MD in each group. Longitudinal studies tracking changes in FA/MD values in athletes at various stages of individual skill development will help clarify whether individuals who go on to become elite athletes possess specific characteristics prior to formal training. In addition, the present study used a relatively small sample size, and therefore, we did not analyze the correlations between fitness data and DTI measures or pursue the DTI analyses using a whole-brain approach. However, our study does provide the basis for future experiments involving larger sample sizes with which to conduct these analyses. Finally, the lack of significant differences between the running and martial arts groups could be because both groups fall on the closed end of the "closed–opened" motor skill continuum. Specifically, both types of athletes developed skills that generally require rhythmic repetitions of the same movements (i.e., closed motor skill). Future studies comparing FA/MD values in athletes with a greater variety of specialized motor skills along the openness continuum (e.g., tennis) should shed light on this issue.

## Conclusion

The present study extends our knowledge of the relationship between athletic expertise and brain plasticity. Our

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results show that both classes of professional athletes had lower FA and higher MD values in the globus pallidus compared with controls, suggesting an association between individual specialized expertise and specific regions of the basal ganglia. These positive modifications may have been caused by increased electrical activity, myelinogenesis, and/or the release of neurotransmitters and neural growth factors due to intensive training, or perhaps even from the increased bending or twisting of neural fibers. Importantly, we found that these changes in basal ganglia plasticity in the white matter occurred regardless of the type of motor skills or sport training involved. Although studies on sport-related neurofibril plasticity are in the early stages, this study contributes to our understanding of the role that the basal ganglia play in this field, and it points the way to future studies on the relationship between athletic experience and white matter integrity in the basal ganglia.

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